Lecture 04: Continuous Probability Distributions

INF1004 Mathematics II

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1 Introduction

In our previous study of discrete random variables, we dealt with countable outcomes, such as the number of heads in a series of coin flips. However, many real-world phenomena—such as time, weight, or distance—exist in a continuous space where an infinite number of values are possible within any given range.

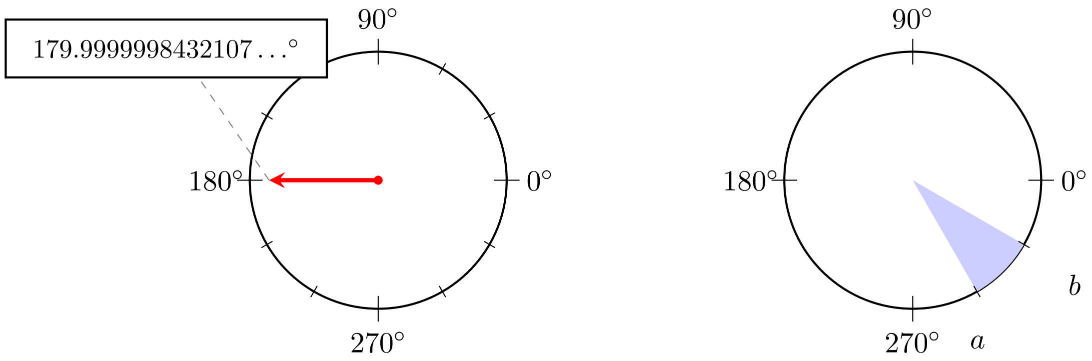

Figure 1: Infinite precision in a continuous space.

A fundamental, yet counter-intuitive, property of continuous spaces is that the probability of a random variable taking an exact, precise value is zero. Consider a spinning arrow on a circular board marked from 0 to 360 degrees. While the arrow must stop at some point, the probability that it stops at exactly 179.9999998° (to infinite precision) is effectively zero. Because there are infinitely many points on the circle, the likelihood of hitting one specific point is negligible. Consequently, in continuous probability, we shift our focus from individual points to the probability of the variable falling within a specific interval.

2 The Probability Density Function (PDF)

To describe the behavior of a continuous random variable X, we use a Probability Density Function, denoted as $f(x)$. Unlike the probability mass function of a discrete variable, $f(x)$ does not represent a probability directly; instead, it represents the "density" of the probability at a specific point.

The probability that X falls within the interval $[a, b]$ is defined as the area under the curve of $f(x)$ between those two points. Mathematically, this is expressed using the integral:

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx \tag{1}$$

For a function to be a valid PDF, it must satisfy two primary conditions: the density must be non-negative for all values ($f(x) \geq 0$), and the total area under the entire curve must equal exactly one:

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \tag{2}$$

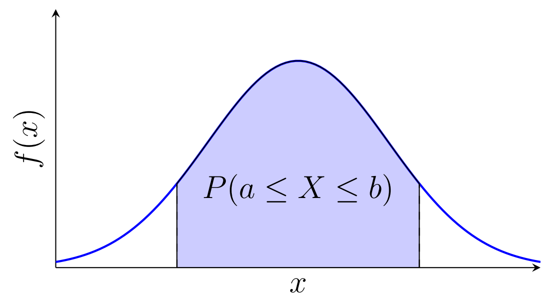

Figure 2: Probability as the area under the density curve.

3 The Continuous Uniform Distribution

The simplest continuous distribution is the Uniform Distribution, often referred to as the rectangular distribution. This model is used when every interval of the same length within the distribution's support is equally probable. If a variable X is uniformly distributed over the interval $[a, b]$, its density function is constant:

$$f(x) = \frac{1}{b-a} \quad \text{for } a \leq x \leq b \tag{3}$$

Returning to our spinning arrow example (from 0° to 360°), the density is $f(x) = 1/360$. The probability of the arrow stopping between 260° and 290° is simply the ratio of the width of that interval to the total width:

$$P(260 \leq X \leq 290) = \frac{290-260}{360} = \frac{30}{360} = \frac{1}{12} \tag{4}$$

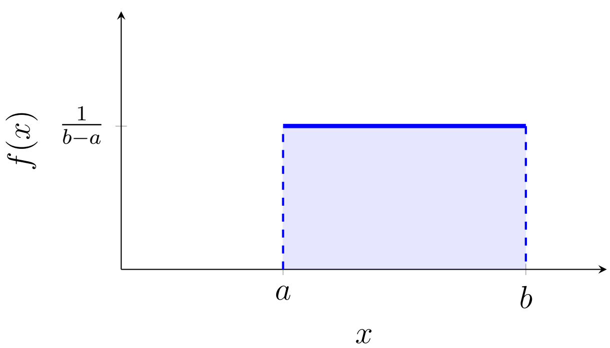

Figure 3: Uniform distribution over an interval $[a, b]$.

4 The Normal Distribution

The Normal Distribution, characterized by its iconic bell-shaped curve, is perhaps the most significant distribution in statistics due to its ability to model many natural phenomena. It is defined by two parameters: the mean (μ), which determines the center of the distribution, and the variance (σ^2), which determines its spread. We denote this as $X \sim N(\mu, \sigma^2)$.

Because the mathematical formula for the normal density is complex, we often simplify problems by transforming the general normal variable X into the Standard Normal Distribution, denoted as Z, where the mean is 0 and the variance is 1 ($Z \sim N(0, 1)$). This transformation is achieved using the Z-score formula:

$$Z = \frac{X - \mu}{\sigma} \tag{5}$$

By converting to Z, we can use standardized Z-tables to find probabilities. The symmetry of the normal curve allows us to derive various probabilities easily; for instance, $P(Z > z) = 1 - P(Z < z)$, and due to its symmetry around zero, $P(Z < -z) = P(Z > z)$.

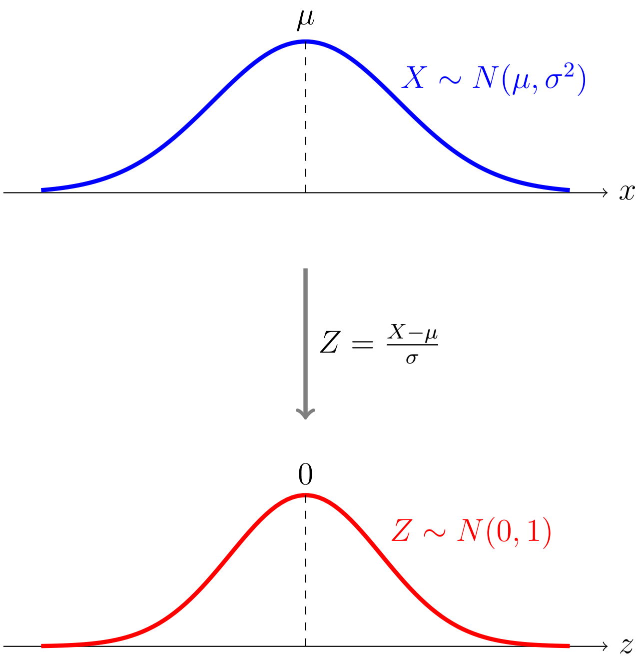

Figure 4: Standardization of a Normal Distribution.

5 Normal Approximation of the Binomial Distribution

Calculating binomial probabilities for large sample sizes (n) can be computationally exhausting. Fortunately, when certain conditions are met—specifically when $np > 5$ and $n(1-p) > 5$—the discrete Binomial Distribution $B(n, p)$ can be closely approximated by a Normal Distribution with $\mu = np$ and $\sigma^2 = np(1-p)$.

When performing this approximation, we must apply a "continuity correction." Since we are using a continuous curve to model discrete steps, we adjust the boundaries by 0.5. For example, to find the probability that X is greater than 25 in a discrete sense, we calculate the area under the normal curve for $Y > 25.5$. This adjustment accounts for the fact that the discrete value "25" actually occupies the space from 24.5 to 25.5 in a continuous model.

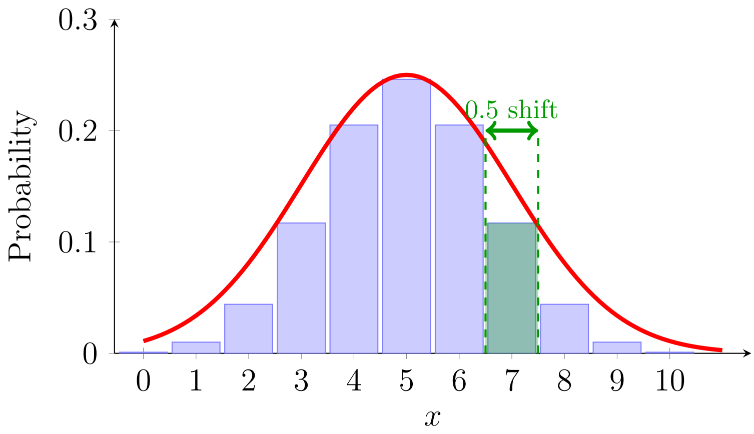

Figure 5: Fitting a normal curve to a binomial distribution.

6 Combining Random Variables

In many statistical applications, we need to consider the sum or difference of multiple random variables. If we have two independent variables X and Y, the expected value of their sum or difference is simply the sum or difference of their individual expectations:

$$E(aX \pm bY) = aE(X) \pm bE(Y) \tag{6}$$

However, the rule for variance is different. Even when subtracting variables, the uncertainties (variances) always add, provided the variables are independent:

$$Var(aX \pm bY) = a^2 Var(X) + b^2 Var(Y) \tag{7}$$

If the original variables are normally distributed, their linear combination will also follow a normal distribution, allowing us to calculate probabilities for the combined outcome using the same Z-score techniques discussed earlier.

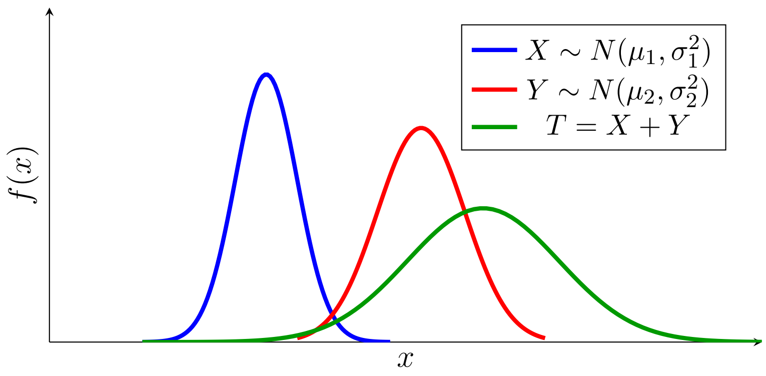

Figure 6: Sum of independent normal random variables.

7 Statistical Studies and the Logic of Inference

The ultimate goal of studying these distributions is to perform statistical inference—using data from a small sample to draw conclusions about a larger population. A population parameter (like the true mean μ) is often unknown, so we calculate a sample statistic (like the sample mean $\bar{x}$) to estimate it.

The validity of these inferences depends heavily on the study design. In an **Observational Study**, researchers simply observe and measure participants without intervention. While these studies are excellent for establishing **correlation** (showing that two variables move together), they cannot establish **causation** due to the potential influence of "lurking variables." To establish a cause-and-effect relationship, a **Randomized Controlled Experiment** is required, where participants are randomly assigned to a treatment or control group to isolate the effect of the variable being studied.

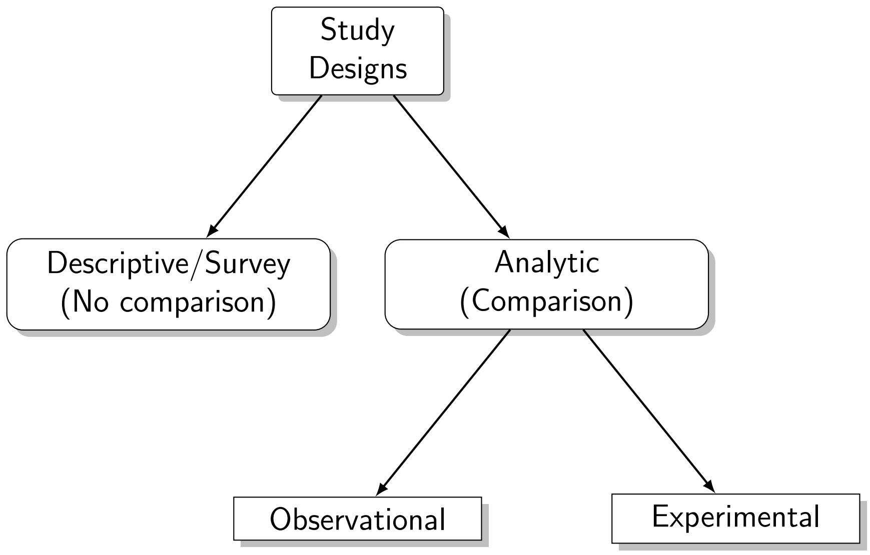

Figure 7: Categorization of statistical studies.